

superscripts:

$$2x^{34}$$
$$2x^{3x^4+5}$$

subscripts:

$$x_1$$
$$x_{12}$$
$$x_{123}$$

greek letter:

$$\pi$$
$$\alpha$$
$$A=\pi r^2$$

trig functions:

$$y=\sin x$$

log functions:

$$\log x$$
$$\ln x$$
$$\log_5 x$$

square roots:

$$\sqrt{2}$$
$$\sqrt[3]{2}$$
$$\sqrt{x^2+y^2}$$
$$\sqrt{1+\sqrt{x}}$$

fractions:

About $\frac{2}{3}$ of the glass is full

$$\frac{x}{x^2+x+1}$$
$$\frac{\sqrt{x+1}}{\sqrt{x}-1}$$
$$\frac{1}{1+\frac{1}{x}}$$
$$\sqrt{\frac{x}{x^2+x+1}}$$